

HW 5 - Zeshui Song

- 7.4** 4. A capacitor is constructed from two brass plates, each measuring $2.5 \text{ mm} \times 2.5 \text{ mm}$ and $300 \mu\text{m}$ thick. The two plates are placed such that they face each other and are separated by a $25 \mu\text{m}$ gap. Calculate the resulting capacitance if (a) the intervening dielectric has a permittivity of $15\epsilon_0$; (b) the intervening dielectric has a permittivity of $1.5\epsilon_0$; (c) the plate separation is doubled and the gap is filled with air; (d) the plate area is doubled and the gap is filled with air.

The capacitance of a parallel-plate capacitor is given by the formula:

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

Where $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$ is the permittivity of free space, ϵ_r is the relative permittivity of the dielectric material, A is the area of the plates, and d is the distance between the plates.

- (a) $\epsilon_r = 15$, $A = 6.25 \text{ mm}^2$, $d = 25 \mu\text{m}$

$$C = \frac{15\epsilon_0 A}{d} = \frac{15[8.854 \times 10^{-12} \text{ F/m}] \left[6.25 \text{ mm}^2 \cdot \frac{(10^{-3} \text{ m})^2}{1 \text{ mm}^2} \right]}{[25 \times 10^{-6} \text{ m}]} = \boxed{3.3202 \times 10^{-11} \text{ F}}$$

- (b) $\epsilon_r = 1.5$, $A = 6.25 \text{ mm}^2$, $d = 25 \mu\text{m}$

$$C = \frac{1.5\epsilon_0 A}{d} = \frac{1.5[8.854 \times 10^{-12} \text{ F/m}] \left[6.25 \text{ mm}^2 \cdot \frac{(10^{-3} \text{ m})^2}{1 \text{ mm}^2} \right]}{[25 \times 10^{-6} \text{ m}]} = \boxed{3.3202 \times 10^{-12} \text{ F}}$$

- (c) $\epsilon_r = 1$ (**air**), $A = 6.25 \text{ mm}^2$, $d = 50 \mu\text{m}$

$$C = \frac{\epsilon_0 A}{d} = \frac{[8.854 \times 10^{-12} \text{ F/m}] \left[6.25 \text{ mm}^2 \cdot \frac{(10^{-3} \text{ m})^2}{1 \text{ mm}^2} \right]}{[50 \times 10^{-6} \text{ m}]} = \boxed{1.1067 \times 10^{-12} \text{ F}}$$

- (d) $\epsilon_r = 1$ (**air**), $A = 12.5 \text{ mm}^2$, $d = 25 \mu\text{m}$

$$C = \frac{\epsilon_0 A}{d} = \frac{[8.854 \times 10^{-12} \text{ F/m}] \left[12.5 \text{ mm}^2 \cdot \frac{(10^{-3} \text{ m})^2}{1 \text{ mm}^2} \right]}{[25 \times 10^{-6} \text{ m}]} = \boxed{4.427 \times 10^{-12} \text{ F}}$$

7.9

9. A silicon *pn* junction diode is characterized by a junction capacitance C_j defined as

$$C_j = \frac{K_s \epsilon_0 A}{W}$$

where $K_s = 11.8$ for silicon, ϵ_0 is the vacuum permittivity, A = the cross-sectional area of the junction, and W is known as the depletion width of the junction. Width W depends not only on how the diode is fabricated, but also on the voltage applied to its two terminals. It can be computed using

$$W = \sqrt{\frac{2K_s \epsilon_0}{qN}(V_{bi} - V_A)}$$

Thus, diodes are often used in electronic circuits since they can be viewed in this context as voltage-controlled capacitors. Assuming parameter values of $N = 5.0 \times 10^{18} \text{ cm}^{-3}$, $V_{bi} = 0.62 \text{ V}$, and using $q = 1.6 \times 10^{-19} \text{ C}$, calculate the capacitance of a diode with cross-sectional area $A = 2 \mu\text{m}^2$ at applied voltages of $V_A = -1, -3$, and -10 V .

Rewriting the formula for capacitance:

$$C_j = K_s \epsilon_0 A \sqrt{\frac{qN}{2K_s \epsilon_0 (V_{bi} - V_A)}} = A \sqrt{\frac{K_s^2 \epsilon_0^2 qN}{2K_s \epsilon_0 (V_{bi} - V_A)}} = A \sqrt{\frac{K_s \epsilon_0 qN}{2(V_{bi} - V_A)}}$$

$$C_j = \left[2 \mu\text{m}^2 \cdot \frac{(10^{-6} \text{ m})^2}{1 \mu\text{m}^2} \right] \sqrt{\frac{[11.8][8.854 \times 10^{-12} \text{ F/m}][1.6 \times 10^{-19} \text{ C}] \left[5 \times 10^{18} \text{ cm}^{-3} \cdot \frac{(10^{-2} \text{ m})^{-3}}{1 \text{ cm}^{-3}} \right]}{2([0.62 \text{ V}] - V_A)}}$$

$$C_j = (2 \times 10^{-12}) \sqrt{\frac{8.3581 \times 10^{-5}}{2([0.62 \text{ V}] - V_A)}} [\text{F}]$$

Note: $1 \text{ F} = 1 \text{ C/V}$.

(a) $V_A = -1 \text{ V}$

$$C_j = (2 \times 10^{-12}) \sqrt{\frac{8.3581 \times 10^{-5}}{2([0.62 - (-1) \text{ V}]})} = \boxed{1.0158 \times 10^{-14} \text{ F}}$$

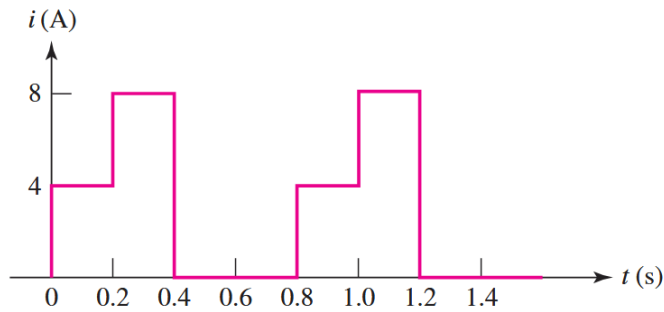
(b) $V_A = -3 \text{ V}$

$$C_j = (2 \times 10^{-12}) \sqrt{\frac{8.3581 \times 10^{-5}}{2([0.62 - (-3) \text{ V}]})} = \boxed{6.7954 \times 10^{-15} \text{ F}}$$

(c) $V_A = -10 \text{ V}$

$$C_j = (2 \times 10^{-12}) \sqrt{\frac{8.3581 \times 10^{-5}}{2([0.62 - (-10) \text{ V}]})} = \boxed{3.9674 \times 10^{-15} \text{ F}}$$

- 7.11** 11. The current flowing through a 1 mF capacitor is shown graphically in Fig. 7.44.
 (a) Assuming the passive sign convention, sketch the resulting voltage waveform across the device. (b) Compute the voltage at 200 ms, 600 ms, and 1.2 s.



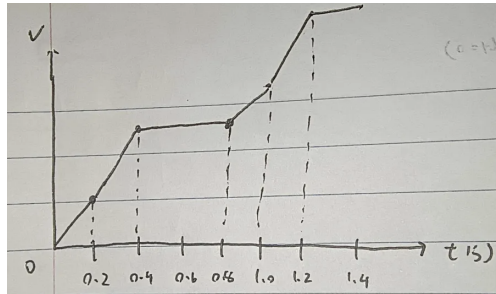
■ **FIGURE 7.44**

The voltage over a capacitor is given by the formula:

$$V = \frac{Q}{C} = \frac{\int i(t) dt}{C} = \frac{i}{C} t$$

Note: Passive sign convention means that the capacitor is absorbing current (charging), so in the context of KVL, there will be a voltage drop in the direction of the current.

(a) Since the current is constant at varying amplitudes, the voltage across the capacitor will increase with different slopes (higher current will lead to a steeper slope).



(b) We can sum up the charge accumulated in the capacitor at each time interval to find the voltage across the capacitor at each time.

$$V_{t=0 \rightarrow 0.2} = \frac{[4 \text{ A}][0.2 \text{ s}]}{[10^{-3} \text{ F}]} = \boxed{800 \text{ V} \quad (\text{at } 200 \text{ ms})}$$

$$V_{t=0 \rightarrow 0.4} = V_{t=0 \rightarrow 0.2} + \frac{[8 \text{ A}][0.2 \text{ s}]}{[10^{-3} \text{ F}]} = \boxed{2400 \text{ V} \quad (\text{at } 600 \text{ ms})}$$

$$V_{t=0 \rightarrow 1.2} = V_{t=0 \rightarrow 0.4} + \frac{[4 \text{ A}][0.2 \text{ s}]}{[10^{-3} \text{ F}]} + \frac{[8 \text{ A}][0.2 \text{ s}]}{[10^{-3} \text{ F}]} = \boxed{4800 \text{ V} \quad (\text{at } 1.2 \text{ s})}$$

- 7.13** 13. A 150 pF capacitor is connected to a voltage source such that $v_C(t) = 12e^{-2t}$ V, $t \geq 0$ and $v_C(t) = 12$ V, $t < 0$. Calculate the energy stored in the capacitor at t equal to (a) 0; (b) 200 ms; (c) 500 ms; (d) 1 s.

The energy stored in a capacitor is given by the formula:

$$w = \frac{1}{2}CV^2(t) = \frac{q^2(t)}{2C}$$

(a) at $t = 0$ s

$$w = \frac{1}{2}[150 \times 10^{-12} \text{ F}](12e^{-2})^2 = \boxed{10.8 \times 10^{-9} \text{ J}}$$

(b) at $t = 0.2$ s

$$w = \frac{1}{2}[150 \times 10^{-12} \text{ F}](12e^{-0.4})^2 = \boxed{4.852 \times 10^{-9} \text{ J}}$$

(c) at $t = 0.5$ s

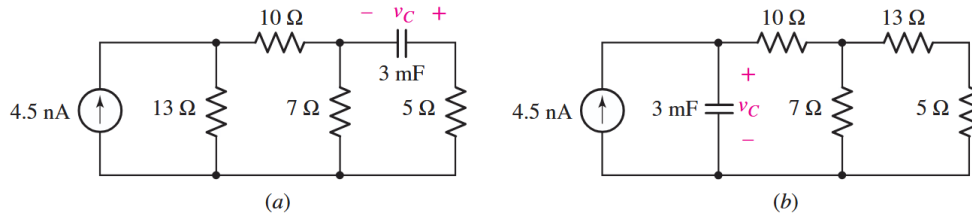
$$w = \frac{1}{2}[150 \times 10^{-12} \text{ F}](12e^{-1})^2 = \boxed{1.4616 \times 10^{-9} \text{ J}}$$

(d) at $t = 1$ s

$$w = \frac{1}{2}[150 \times 10^{-12} \text{ F}](12e^{-2})^2 = \boxed{0.1978 \times 10^{-9} \text{ J}}$$

7.15

15. For each circuit shown in Fig. 7.46, calculate the voltage labeled v_C .

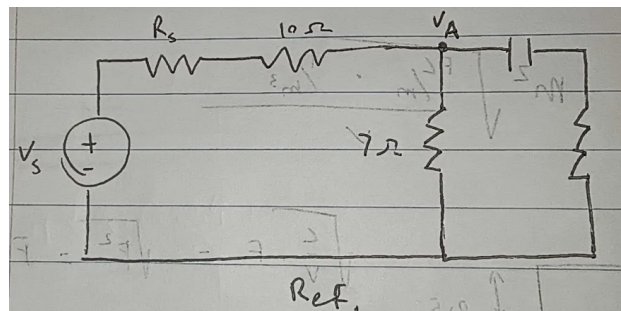


Assuming that the circuit is in steady state, the capacitor will act like an open circuit, with no current flowing through it.

(a) Start with a source transformation of the current source to a voltage source:

$$V_s = i_s R_s = [4.5 \times 10^{-9} \text{ A}][13 \Omega] = 5.85 \times 10^{-8} \text{ V}$$

$$R_s = R_p = 13 \Omega$$



Thus, the current through the capacitor will be equal to the voltage over the $7\ \Omega$ resistor. We can find it using voltage division:

Combine the R_s and $10\ \Omega$ resistor in series:

$$R = 13 + 10\ \Omega = 23\ \Omega$$

$$V_{7\Omega} = [5.85 \times 10^{-8}\text{ V}] \cdot \frac{7\ \Omega}{7\ \Omega + 23\ \Omega} = 1.365 \times 10^{-8}\text{ V}$$

Since the diagram shows the negative terminal of V_c is equal to the positive terminal of $V_{7\Omega}$, the voltage across the capacitor has to be negative. This also makes sense since it is resisting the current flow from the voltage source, so it will be charged in the opposite direction of the voltage source.

$$\Rightarrow \boxed{V_c = -1.365 \times 10^{-8}\text{ V}}$$

(b) The voltage over the capacitor is equal to the voltage over the current source. Thus, we start by finding the equivalent resistance of the circuit:

Combine $13\ \Omega$ and $5\ \Omega$ in series:

$$R_1 = 13 + 5\ \Omega = 18\ \Omega$$

Combine $R_1 = 18\ \Omega$ and $7\ \Omega$ in parallel:

$$R_2 = \frac{18 \cdot 7}{18 + 7}\ \Omega = 5.04\ \Omega$$

Combine $R_2 = 5.04\ \Omega$ and $10\ \Omega$ in series:

$$R_{eq} = 5.04 + 10\ \Omega = 15.04\ \Omega$$

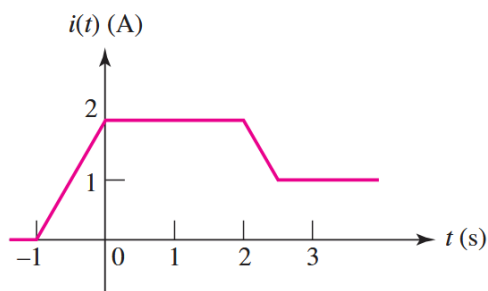
Thus, the voltage across the current source is:

$$V_s = i_s R_{eq} = [4.5 \times 10^{-9}\text{ A}][15.04\ \Omega] = 6.768 \times 10^{-8}\text{ V}$$

Based on the polarity of the capacitor:

$$\Rightarrow \boxed{V_c = 6.768 \times 10^{-8}\text{ V}}$$

- 7.17** 17. If the current flowing through a 75 mH inductor has the waveform shown in Fig. 7.47, (a) sketch the voltage which develops across the inductor terminals for $t \geq 0$, assuming the passive sign convention; and (b) calculate the voltage at $t = 1\text{ s}$, 2.9 s , and 3.1 s .



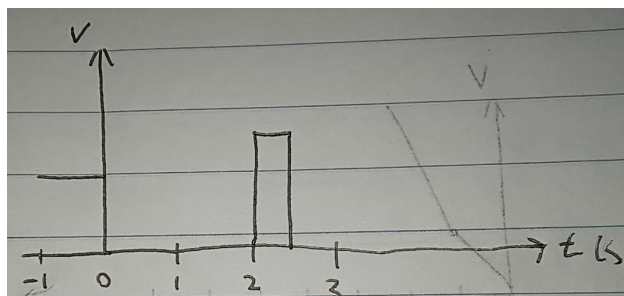
■ FIGURE 7.47

The voltage over an inductor is given by the formula:

$$V = L \frac{di}{dt}$$

Since the slopes of the current time plot are constant, the voltage across the inductor will be a step function, with the voltage being positive when the current is increasing and negative when the current is decreasing.

(a)



(b)

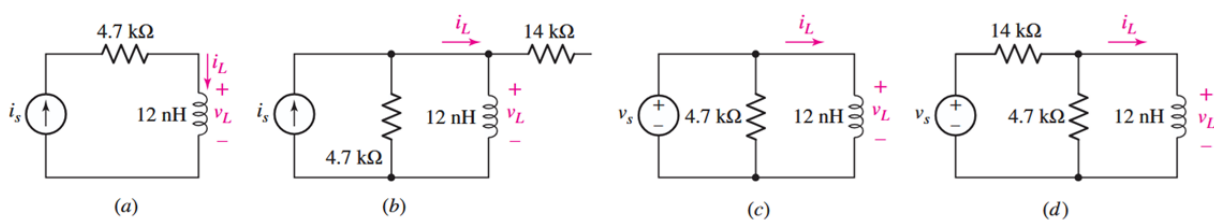
$$V(t = 1) = L \left. \frac{di}{dt} \right|_{t=1} = [75 \times 10^{-3} \text{ H}](0) = \boxed{0 \text{ V}}$$

$$V(t = 2.9) = L \left. \frac{di}{dt} \right|_{t=2.9} = [75 \times 10^{-3} \text{ H}](0) = \boxed{0 \text{ V}}$$

$$V(t = 3.1) = L \left. \frac{di}{dt} \right|_{t=3.1} = [75 \times 10^{-3} \text{ H}](0) = \boxed{0 \text{ V}}$$

7.21

21. Calculate v_L and i_L for each of the circuits depicted in Fig. 7.49 if $i_s = 1 \text{ mA}$ and $v_s = 2 \text{ V}$.



■ FIGURE 7.49

The voltage over an inductor is given by the formula:

$$V = L \frac{di}{dt}$$

Note that because the current in each circuit is constant, $\frac{di}{dt} = 0$, so the voltage across the inductor will be zero in all cases. This means that the inductor will act like a short circuit, and the voltage across it will be zero.

(a)

$$V_L = 0 \text{ V}$$

$$i_L = i_s = 1 \text{ mA}$$

(b)

$$V_L = 0 \text{ V}$$

The inductor acts like a short, so all the current will flow through it instead (path of least resistance):

$$i_L = i_s = 1 \text{ mA}$$

(c)

$$V_L = 0 \text{ V}$$

Beacuse we basically shorted the voltage source, the current will be infinite across the inductor:

$$i_L = \infty \text{ mA}$$

(d)

$$V_L = 0 \text{ V}$$

Using Ohm's law, we can find the current in the circuit. The $14 \text{ k}\Omega$ resistor has a voltage drop equal to the voltage source, because no current flows through the $4.7 \text{ k}\Omega$ resistor since the inductor is a short circuit:

$$v_s = i_L [14 \text{ k}\Omega] \implies i_L = \frac{v_s}{14 \text{ k}\Omega} = \frac{2 \text{ V}}{14 \times 10^3 \Omega} = 1.428 \times 10^{-4} \text{ A}$$

$$i_L = 0.1428 \text{ mA}$$

7.26 26. Calculate the energy stored in a 1 nH inductor if the current flowing through it is (a) 0 mA ; (b) 1 mA ; (c) 20 A ; (d) $5 \sin 6t \text{ mA}$, $t > 0$.

The energy stored in an inductor is given by the formula:

$$w = \frac{1}{2} L i^2(t)$$

(a) $i = 0 \text{ mA}$

$$w = \frac{1}{2} L i^2(t) = \frac{1}{2} [10^{-9} \text{ H}] [0 \text{ A}]^2 = 0 \text{ J}$$

(b) $i = 1 \text{ mA}$

$$w = \frac{1}{2} L i^2(t) = \frac{1}{2} [10^{-9} \text{ H}] [10^{-3} \text{ A}]^2 = 5 \times 10^{-16} \text{ J}$$

(c) $i = 20 \text{ A}$

$$w = \frac{1}{2} L i^2(t) = \frac{1}{2} [10^{-9} \text{ H}] [20 \text{ A}]^2 = 0.2 \times 10^{-6} \text{ J}$$

(d) $i = 5 \sin(6t) \text{ mA}$

$$w = \frac{1}{2} L i^2(t) = \frac{1}{2} [10^{-9} \text{ H}] [5 \times 10^{-3} \sin(6t) \text{ A}]^2 = 1.25 \times 10^{-14} \sin^2(6t) \text{ J}$$

- 7.28** 28. Making the assumption that the circuits in Fig. 7.51 have been connected for a very long time, determine the value for each current labeled i_x .

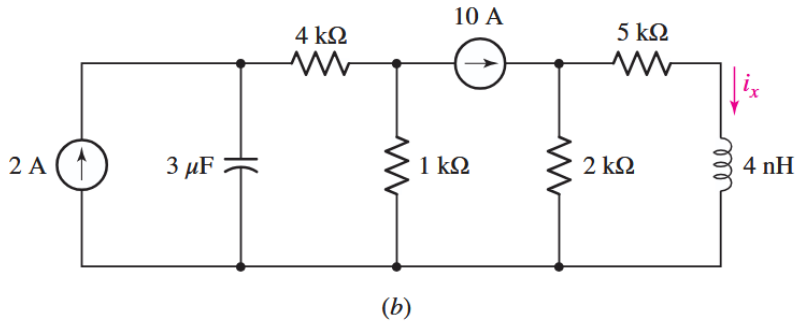
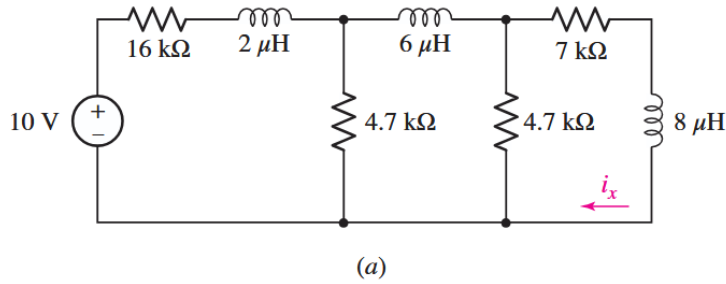


FIGURE 7.51

At steady state, the inductor will act like a short circuit with no resistance, so the voltage across it will be zero.

(a) The current i_x can be found using Ohm's law if I can find the voltage over the 7 kΩ resistor. Since the inductor is like a short circuit (wire), the 7 kΩ and the two 4.7 kΩ resistors are in parallel so:

$$\frac{1}{R_1} = \frac{1}{7 \text{ k}\Omega} + \frac{1}{4.7 \text{ k}\Omega} + \frac{1}{4.7 \text{ k}\Omega} \implies R_1 = 1.759 \text{ k}\Omega$$

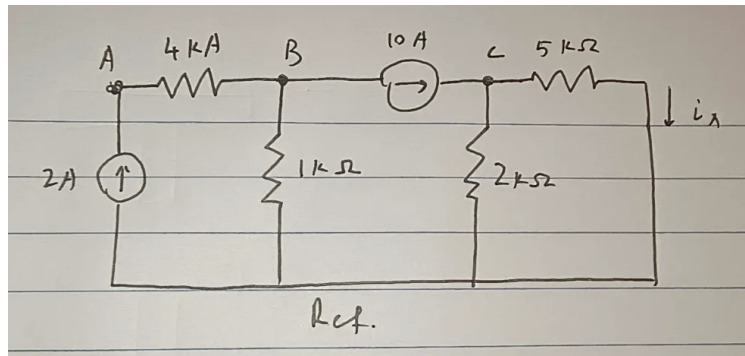
Now we just have resistor R_1 in series with the 16 kΩ resistor, so we can find the voltage over R_1 using voltage division:

$$V_{R_1} = v_s \cdot \frac{R_1}{R_1 + 16 \text{ k}\Omega} = [10 \text{ V}] \cdot \frac{[1.759 \text{ k}\Omega]}{[1.759 + 16 \text{ k}\Omega]} = 0.99066 \text{ V}$$

This voltage is the same as the voltage across the 7 kΩ resistor, since it is in parallel, so we can find the current i_x using Ohm's law:

$$i_x = \frac{V_{R_1}}{7 \text{ k}\Omega} = \frac{[0.99066 \text{ V}]}{[7 \times 10^3 \Omega]} = \boxed{1.415 \times 10^{-4} \text{ A}}$$

(b) At steady state, the capacitor becomes an open circuit as well. So we can redraw the circuit:



Using nodal analysis to find the voltage over the $5\text{ k}\Omega$ resistor, we can find the current i_y using Ohm's law.

KCL at node A:

$$[2\text{ A}] = \frac{V_A - V_B}{[4 \times 10^3 \Omega]}$$

KCL at node B:

$$\frac{V_A - V_B}{[4 \times 10^3 \Omega]} = [10\text{ A}] + \frac{V_B}{[10^3 \Omega]}$$

KCL at node C:

$$[10\text{ A}] = \frac{V_C}{[2 \times 10^3 \Omega]} + \frac{V_C}{[5 \times 10^3 \Omega]}$$

$$\Rightarrow V_C = 14485.714\text{ V}$$

Using Ohm's law, we can find the current i_x :

$$i_y = \frac{V_C}{[5 \times 10^3 \Omega]} = \frac{[14485.714\text{ V}]}{[5 \times 10^3 \Omega]} = \boxed{2.857\text{ A}}$$

7.29

29. Calculate the voltage labeled v_x in Fig. 7.52, assuming the circuit has been running a very long time, if (a) a $10\ \Omega$ resistor is connected between terminals x and y ; (b) a $1\ \text{H}$ inductor is connected between terminals x and y ; (c) a $1\ \text{F}$ capacitor is connected between terminals x and y ; (d) a $4\ \text{H}$ inductor in parallel with a $1\ \Omega$ resistor is connected between terminals x and y .

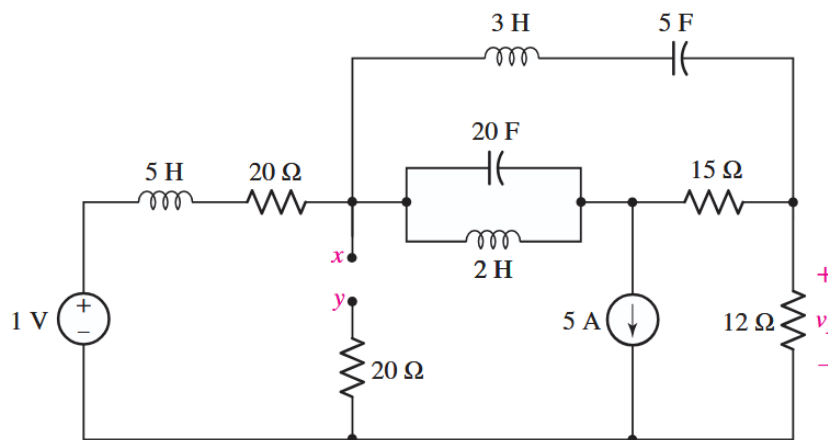
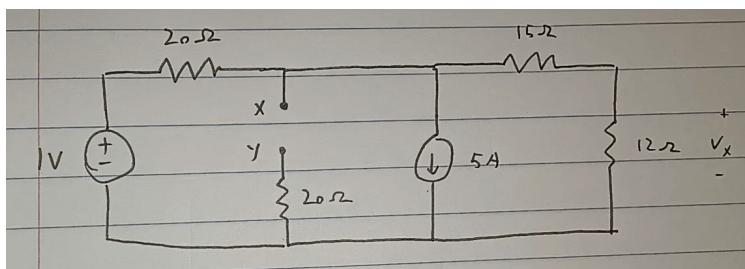
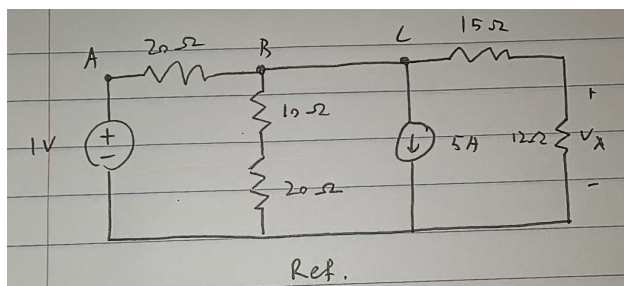


FIGURE 7.52

Redrawing the circuit at steady state:



(a) Using nodal analysis, we can find $V_C = V_B$ and then use voltage division to find the voltage across the $12\ \Omega$ resistor V_x .



Node A:

$$V_A = 1\ \text{V}$$

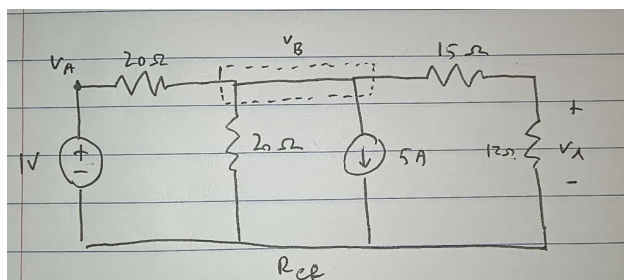
Node B:

$$\begin{aligned} \frac{V_A - V_B}{[20\ \Omega]} &= \frac{V_B}{[10 + 20\ \Omega]} + [5\ \text{A}] + \frac{V_B}{[15 + 12\ \Omega]} \\ \Rightarrow -4.95 &= 0.12037V_B \Rightarrow V_B = -41.123\ \text{V} \end{aligned}$$

Since $V_C = V_B$, we can find the voltage across the $12\ \Omega$ resistor using voltage division:

$$V_x = V_C \cdot \frac{[12\ \Omega]}{[12 + 15\ \Omega]} = [-41.123\ \text{V}] \cdot \frac{[12\ \Omega]}{[27\ \Omega]} = \boxed{-18.276\ \text{V}}$$

(b)



Node A:

$$V_A = 1\ \text{V}$$

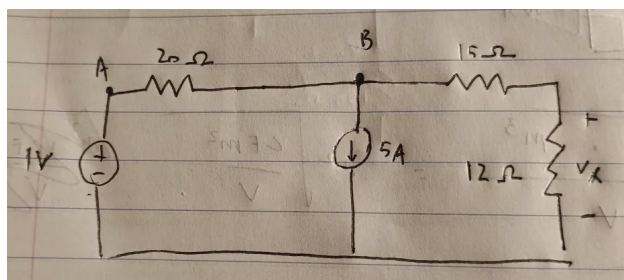
Node B:

$$\begin{aligned} \frac{V_A - V_B}{[20\ \Omega]} &= \frac{V_B}{[20\ \Omega]} + [5\ \text{A}] + \frac{V_B}{[15 + 12\ \Omega]} \\ \Rightarrow -4.95 &= 0.13703V_B \Rightarrow V_B = -36.121\ \text{V} \end{aligned}$$

We can use voltage division to find the voltage across the $12\ \Omega$ resistor:

$$V_x = V_B \cdot \frac{[12\ \Omega]}{[12 + 15\ \Omega]} = [-36.121\ \text{V}] \cdot \frac{[12\ \Omega]}{[27\ \Omega]} = \boxed{-16.053\ \text{V}}$$

(c)



Node A:

$$V_A = 1\ \text{V}$$

Node B:

$$\begin{aligned} \frac{V_A - V_B}{[20\ \Omega]} &= [5\ \text{A}] + \frac{V_B}{[15 + 12\ \Omega]} \\ \Rightarrow -4.95 &= 0.08703V_B \Rightarrow V_B = -56.872\ \text{V} \end{aligned}$$

We can use voltage division to find the voltage across the $12\ \Omega$ resistor:

$$V_x = V_B \cdot \frac{[12\ \Omega]}{[12 + 15\ \Omega]} = [-56.872\ \text{V}] \cdot \frac{[12\ \Omega]}{[27\ \Omega]} = \boxed{-25.276\ \text{V}}$$

(d) Since current flows through path of least resistance, it'll be the same case as part (b).

7.35 35. Determine the equivalent capacitance C_{eq} of the network shown in Fig. 7.57.

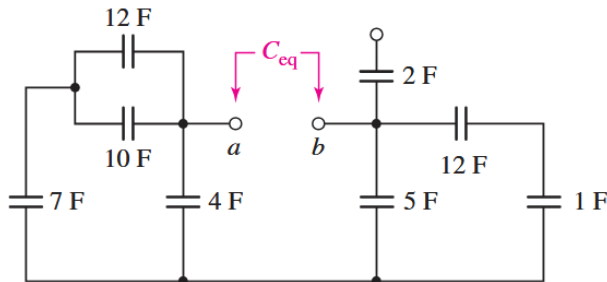


FIGURE 7.57

Capacitors in parallel add like:

$$C_{eq} = C_1 + C_2 + C_3$$

Capacitors in series add like:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Working from the right: Add the 12 F and 1 F capacitors in series:

$$C_1 = \frac{[12 \text{ F}][1 \text{ F}]}{[12 + 1 \text{ F}]} = 0.923 \text{ F}$$

Add the 0.923 F capacitor in parallel with the 5 F capacitor:

$$C_2 = C_1 + 5 = 0.923 + 5 = 5.923 \text{ F}$$

For the left side: add the 12 F and 10 F capacitors in parallel:

$$C_3 = 12 + 10 = 22 \text{ F}$$

Add C_3 and 7 F in series:

$$C_4 = \frac{[22 \text{ F}][7 \text{ F}]}{[22 + 7 \text{ F}]} = 5.310 \text{ F}$$

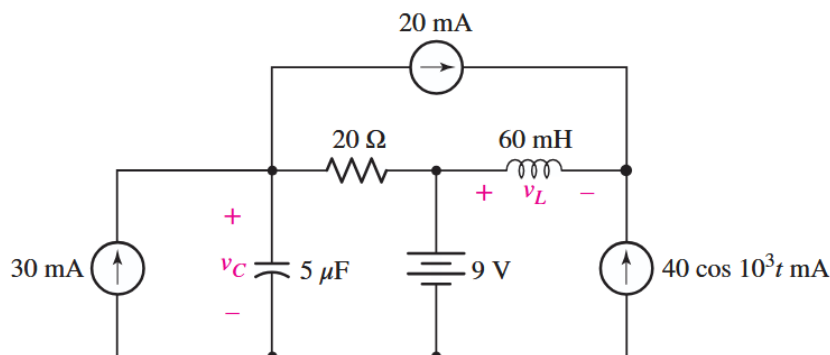
Add C_4 and 4 F in parallel:

$$C_5 = C_4 + 4 = 5.310 + 4 = 9.310 \text{ F}$$

Finally, we have simplified both sides into 2 capacitors in series: C_5 and C_2 :

$$C_{eq} = \frac{[9.310 \text{ F}][5.923 \text{ F}]}{[9.310 + 5.923 \text{ F}]} = \boxed{3.619 \text{ F}}$$

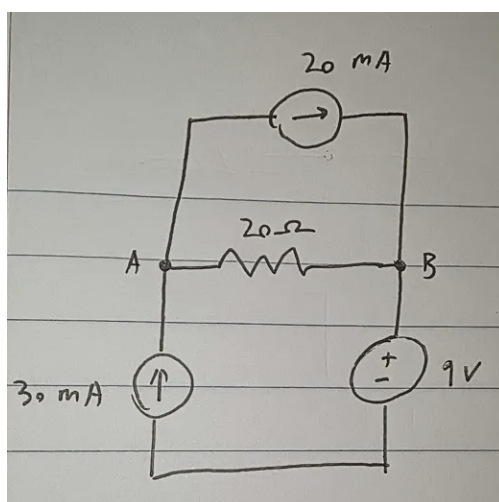
- 7.49** 49. If it is assumed that all the sources in the circuit of Fig. 7.70 have been connected and operating for a very long time, use the superposition principle to find $v_C(t)$ and $v_L(t)$.



■ **FIGURE 7.70**

Using superposition, we can find the voltages across the inductor and capacitor at steady state from the DC sources and AC current source separately, and then add them together to find the total voltage across the inductor and capacitor.

DC sources only: Following superposition, turning off the current source becomes an open circuit. Redrawing the circuit at steady state:



At steady state, $\frac{di}{dt} = 0$ so the inductor has a voltage of

$$\underline{V_{L,DC} = 0 \text{ V}}$$

The capacitor has a voltage equal to V_A . We can find V_A using nodal analysis:

Node A:

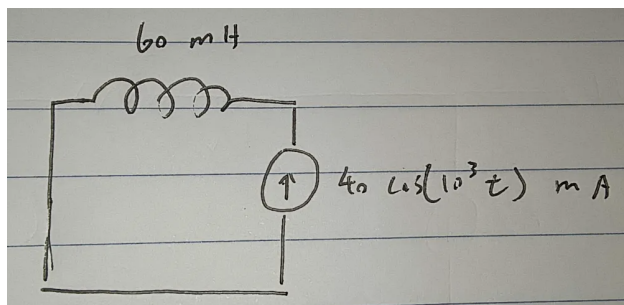
$$[30 \times 10^{-3} \text{ A}] = \frac{V_A - V_B}{[20 \Omega]} + [20 \times 10^{-3} \text{ A}]$$

Node B:

$$V_B = 9 \text{ V}$$

$$\Rightarrow V_A = 9.2 \text{ V} \Rightarrow \underline{V_{C,DC} = 9.2 \text{ V}}$$

AC current source only: Following superposition, turning off the voltage sources becomes short circuits, and current sources becomes open circuits. Redrawing the circuit at steady state:



Since the voltage source becomes a short circuit, the capacitor then has a voltage of zero (no current flows through it):

$$\underline{V_{C,AC} = 0 \text{ V}}$$

For the inductor, we can differentiate the current to find the voltage across it:

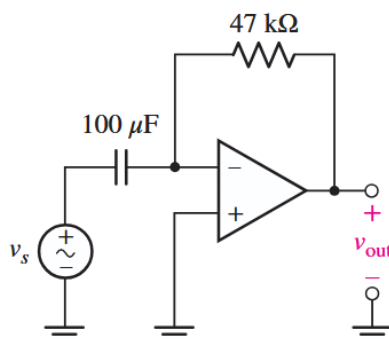
$$V_{L,AC} = L \frac{di}{dt} = [60 \times 10^{-3} \text{ H}] \cdot \frac{d}{dt} [40 \cos(1000t) \times 10^{-3} \text{ A}] = \underline{-2.4 \sin(1000t) \text{ V}}$$

Finally, applying superposition:

$$\boxed{V_C = V_{C,DC} + V_{C,AC} = 9.2 \text{ V} + 0 \text{ V} = 9.2 \text{ V}}$$

$$\boxed{V_L = V_{L,DC} + V_{L,AC} = 0 \text{ V} - 2.4 \sin(1000t) \text{ V} = -2.4 \sin(1000t) \text{ V}}$$

- 7.54** 54. For the circuit shown in Fig. 7.73, assume no energy is initially stored in the capacitor, and determine v_{out} if v_s is given by (a) $5 \sin 20t \text{ mV}$; (b) $2e^{-t} \text{ V}$.



■ FIGURE 7.73

Since the non-inverting terminal of the op amp is grounded, the inverting terminal will also be at 0 V. Thus, the voltage drop across the capacitor must be equal to the voltage source V_s .

$$V_C = V_s - [0 \text{ V}]$$

Applying KCL at the node between the capacitor and the 47 k Ω resistor:

$$i_C = i_R$$

Where i_C is found through differentiating the voltage across the capacitor:

$$\begin{aligned} V_C = \frac{Q}{C} &\implies \frac{dV_C}{dt} = \frac{1}{C} \frac{dQ}{dt} \\ &\implies i_C = C \frac{dV_s}{dt} \end{aligned}$$

i_R is found through Ohm's law:

$$i_R = \frac{0 - V_{\text{out}}}{[47 \times 10^3 \Omega]}$$

Setting them equal to each other:

$$C \frac{dV_s}{dt} = \frac{0 - V_{\text{out}}}{[47 \times 10^3 \Omega]} \implies V_{\text{out}} = -[47 \times 10^3 \Omega][100 \times 10^{-6} \text{ F}] \frac{dV_s}{dt}$$

Simplified:

$$V_{\text{out}} = -4.7 \frac{dV_s}{dt} [\text{V}]$$

(a) $V_s = 5 \sin(20t) \text{ mV}$

$$\frac{dV_s}{dt} = 5 \cdot 20 \cos(20t) \text{ mV/s} = 100 \cos(20t) \times 10^{-3} \text{ V/s}$$

$$V_{\text{out}} = -4.7(0.1 \cos(20t)) = \boxed{-0.47 \cos(20t) \text{ V}}$$

(b) $V_s = 2e^{-t} \text{ V}$

$$\frac{dV_s}{dt} = -2e^{-t} \text{ V/s}$$

$$V_{\text{out}} = -4.7(-2e^{-t}) = \boxed{9.4e^{-t} \text{ V}}$$